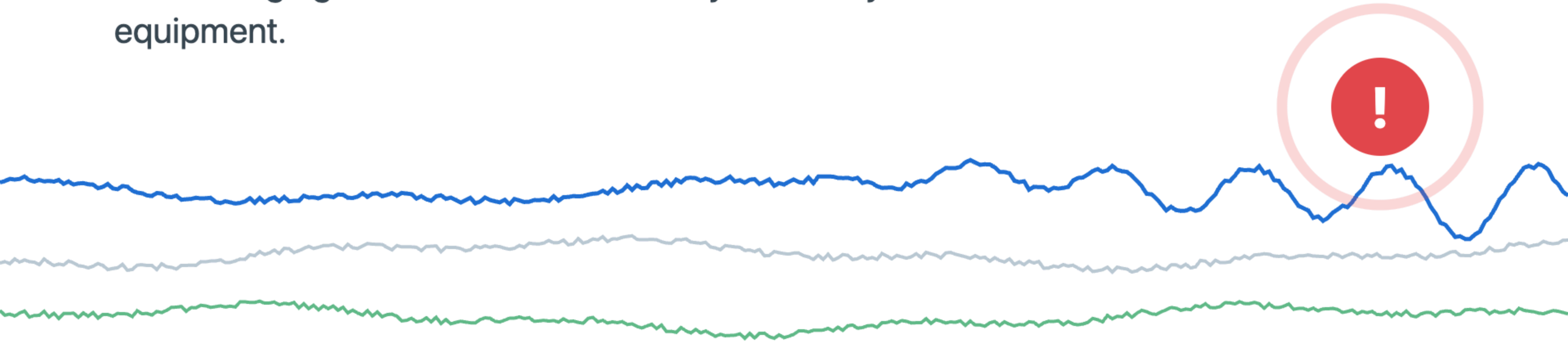
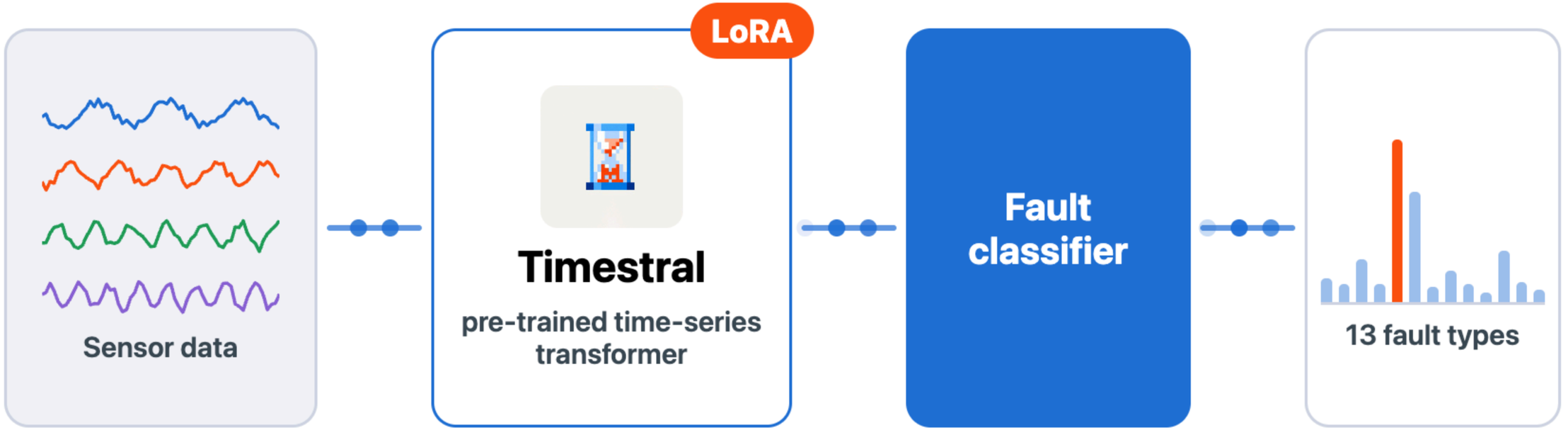
Team ASD · Business Turku challenge

Adding condition awareness to existing machines

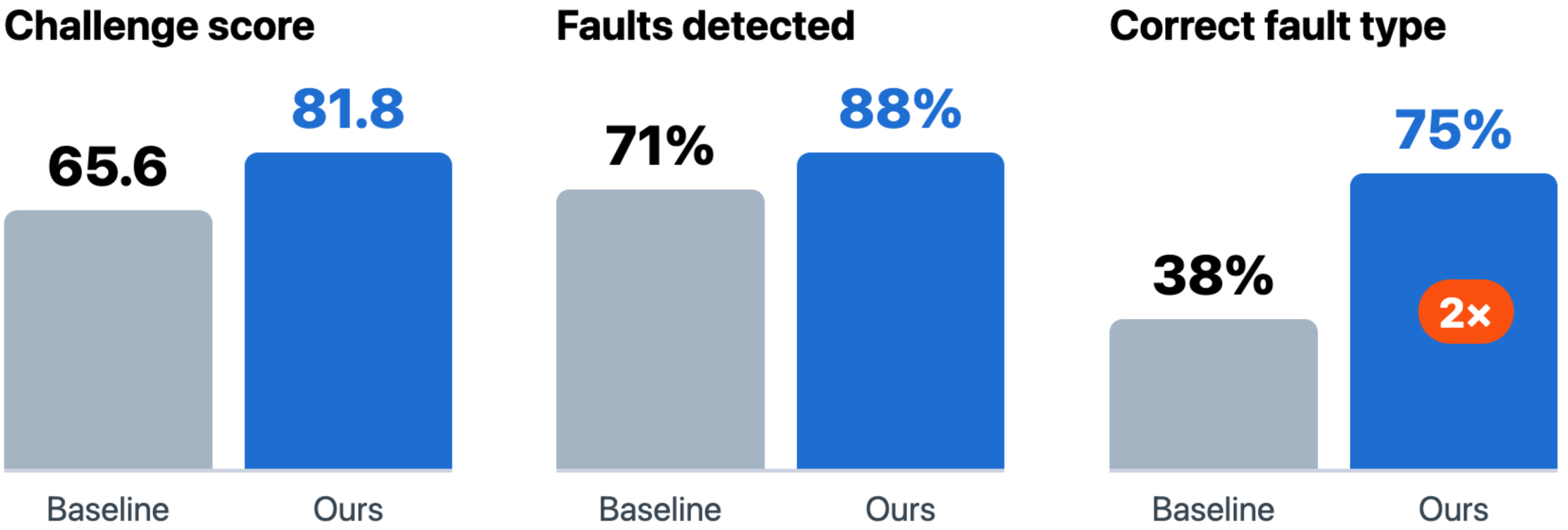
Find emerging faults in the sensor data you already have. No new equipment.



Approach

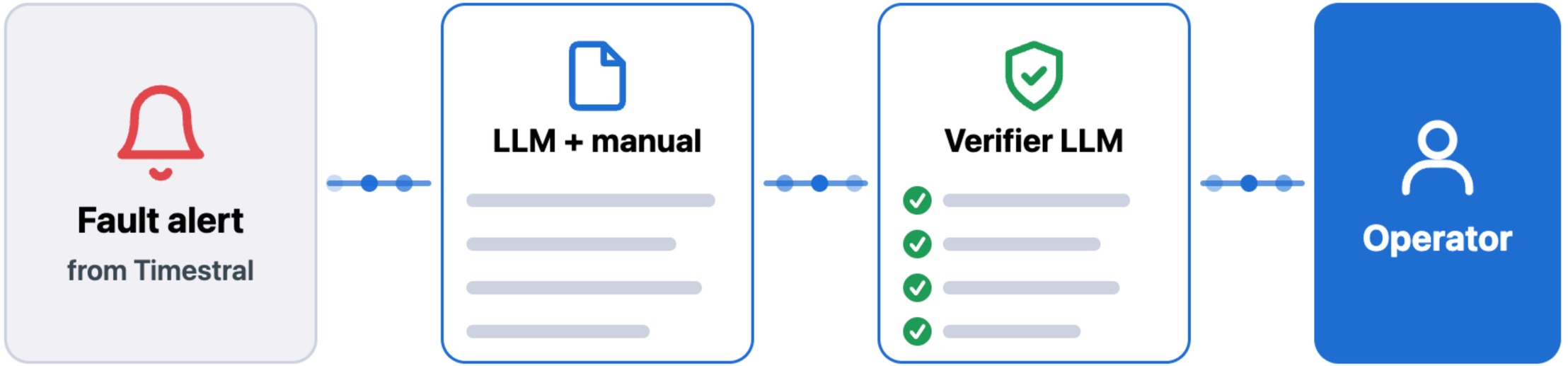

LoRA
Timestral
pre-trained time-series transformer
Fault classifier
Sensor data
13 fault types

Benchmark



7 validation pumps (2 official + 5 generated) · organizer scorer · baseline: organizer B1

Use case


Fault alert
from Timestral
LLM + manual
Verifier LLM
Operator

Cost

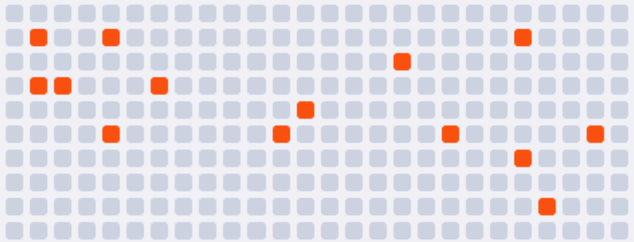
26M

parameters

Runs on-site.
Data stays there.

1.4M

weights trained (of 26M)



$8.97

one-time training

1 × A100
5 hours

Not pump-specific



7 public pumps



+ 250 simulated pump-weeks



Any annotated time series



+ its manual



Same pipeline

Reliability and safety

Alert threshold: set by the customer



Fewer missed alerts

Fewer false alarms

Each alert shows its cause

